

## Sub-Phase 1: Define Problem

|               |                         |
|---------------|-------------------------|
| Date          | 28-09-2026              |
| Team ID       | SWTID-2026-7624 team id |
| Project name  | Script-controlled ACL   |
| Maximum marks | 4 marks                 |

### 1. Problem Statement

In educational institutions and large organizations managing data in ServiceNow, protecting sensitive student and faculty information is a critical requirement. Standard Role-Based Access Control (RBAC) only grants or denies access to an entire table. It lacks the ability to dynamically evaluate field values inside a record, leading to data exposure where users from one department (e.g., ECE or CSE) can view or modify records belonging to another department (e.g., EEE).

### 2. Current Gaps & Challenges

- \* **Lack of Record-Level Security:** Standard role assignments cannot restrict access based on specific field attributes like Branch = EEE.
- \* **Data Privacy Risks:** Sensitive fields like Student Roll Number, Faculty Name, Email, and Phone Number in the u\_institution\_details table are exposed across departments without granular controls.
- \* **Over-Privileged Access:** Without granular CRUD (Create, Read, Update, Delete) separation, users with basic access might inadvertently modify or delete records.

### 3. Root Cause Analysis

Without dynamic scripting in Access Control Lists (ACLs), ServiceNow evaluates security static rules only. To enforce conditional access where record attributes (e.g., Branch) match user authorizations (e.g., bb1 role), a script-controlled ACL mechanism is required.

### 4. Proposed Solution & Goal

Implement a Script-Controlled ACL architecture on the custom u\_institution\_details table:

- \* **Dynamic Read Restriction:** Allow users with the bb1 role to view only records where Branch is set to EEE.
- \* **Granular Operation Rights:** Assign specific custom roles for operational rights:
  - \* bb1 for Read access
  - \* bb2 for Create access
  - \* bb3 for Write/Edit access
  - \* bb4 for Delete access
- \* **Admin Exemption:** Maintain full, unrestricted access for System Administrators (admin role).

### 5. Success Metrics

- \* **100% Data Isolation:** EEE branch users see zero records from ECE or CSE branches.

\* Unauthorized Access Denial: Users lacking designated roles (bb1–bb4) are completely restricted from viewing or modifying records.

\* Verification via Impersonation: Successful verification through ServiceNow user impersonation for both Admin and test user roles.